

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

4th Semester of B. Pharm. Examination

University Theory Examination October 2015

PH212 Physical Pharmaceutics

Date: 16/10/2015, Friday

Time: 01:30 p.m to 4:30 p.m.

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches wherever necessary.

SECTION- I

Q 1 Attempt any TWO of the following

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|----------|---|----------|
| A | 1. Explain characteristics of crystalline & amorphous form | 4 |
| | 2. Write a Note on 'Polymorphism' | 6 |
| B | 1. Write a detail note on Latent heat & Vapour Pressure. | 4 |
| | 2. Briefly explain 'Liquid Crystals' | 6 |
| C | 1. Briefly explain Angle of Repose & Hausner Ratio | 4 |
| | 2. Enlist methods for Particle volume determination. Explain any one. | 6 |

Q 2 Attempt any TWO of the following

- | | | |
|----------|---|----------|
| A | 1. Write a detail note on 'Duonoy Ring Method' | 5 |
| | 2. Write a note on HLB & Required HLB | 5 |
| B | 1. Briefly explain 'Spreading Coefficient' | 5 |
| | 2. Write a Short note on Surfactant | 5 |
| C | 1. Define:- | |
| | a. Buffer Capacity | |
| | b. Isotonic Solution | |
| | c. Paratonic Solution | 5 |
| | d. Hypertonic Solution | |
| | e. Hypotonic Solution | |
| | 2. Write a note on Preparation & Measurements of Isotonic Solution. | 5 |

SECTION- II

Q 3 Attempt any TWO of the following

- A** Enlist the methods for measurement of viscosity. Explain any one in detail **10**
- B** 1. Write a note – Thixotropy... **05**
2. Define : a) Kinematic Viscosity b) Flocculated Suspension **05**
- C** 1. Explain the non-newtonian system. **05**
2. What are the advantages of complexation in pharmacy? **05**

Q 4 Attempt any TWO of the following

- A** 1. Explain in detail Stokes' law for suspension? **05**
2. Explain the methods of identification of emulsions? **05**
- B** 1. Classify the methods for identification of complexation and explain any one. **05**
2. What are application of colloidal system? **05**
- C** 1. Comments:- Viscosity of paracetamol suspension increase, it increase the rate of sedimentation. **10**
2. Comments:- Large particle size of drug is advantages for preparation of emulsion.
3. Comment:- Cyclodextrin is used for solubility enhancements of drug.
4. Comments:- Falling ball instruments is use to measure the flow of suspension.